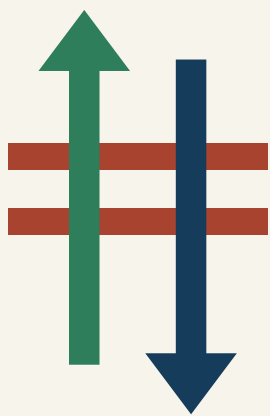


# THE TWO-WAY LINE 京张对开

Running the Centennial Jing-Zhang Corridor in Both Directions — an AI-belt urban design



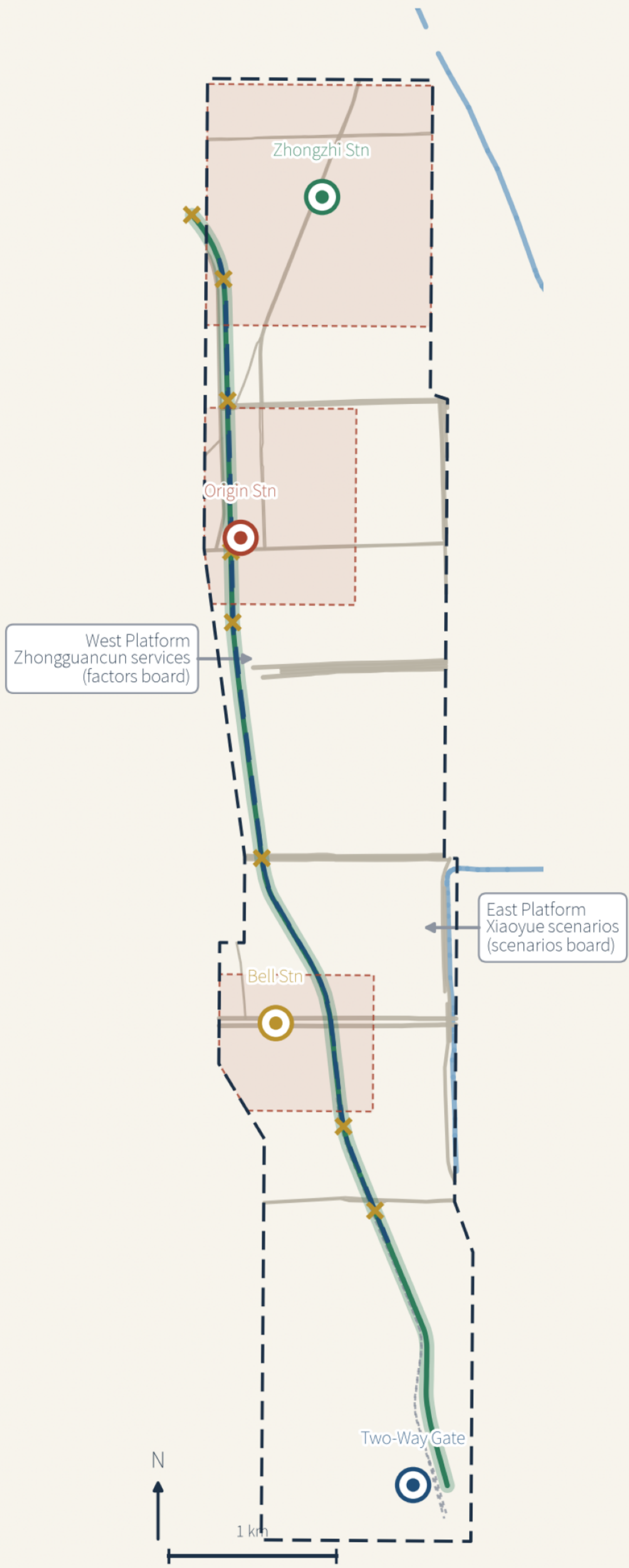
Identity • Two arrows, one track

Two parallel rails bound by sleeper bars, north-pointing above and south-pointing below; the negative space between them forms the skeleton of the character 开 ( "open" ).  
Green = up-line energy and compute, deep blue = the Under Line, brick red = heritage and sleepers.

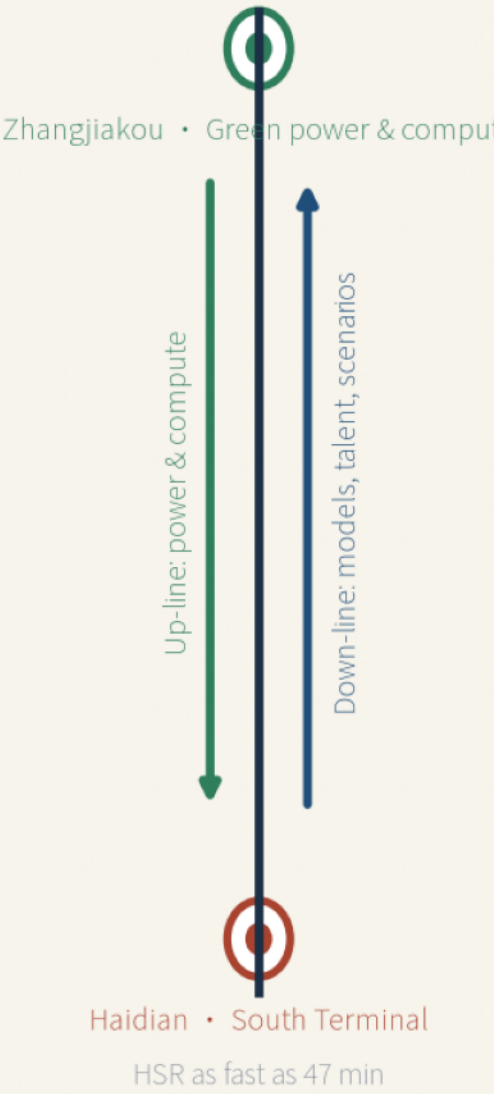
Master Concept Board

## THE TWO-WAY LINE • Master Concept

Surface + Under Lines • 3 Stations 1 Gate • 2 Platforms 8 Crossings • Twin-River Cooling Loop — 11.4 km<sup>2</sup>



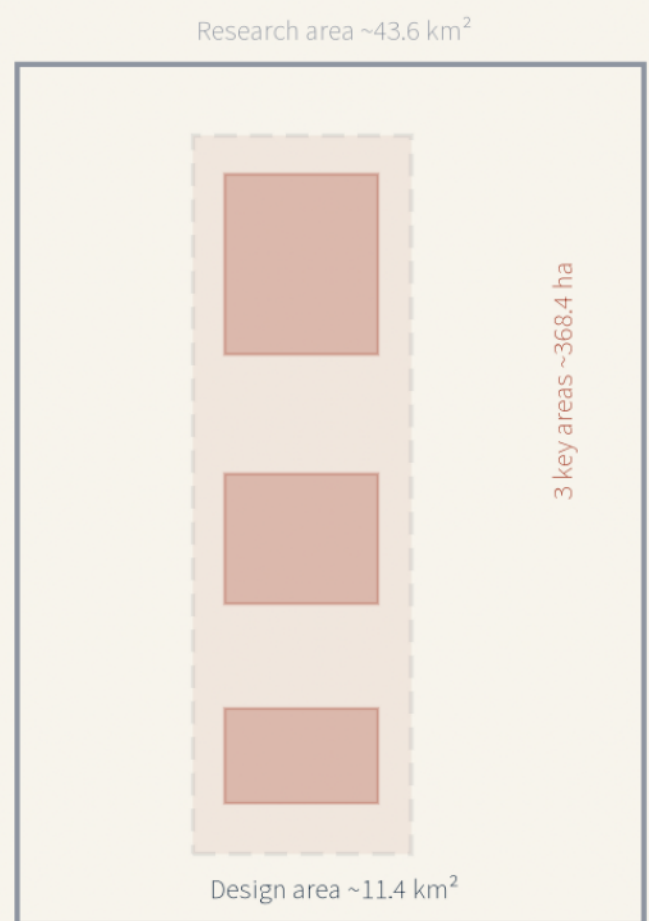
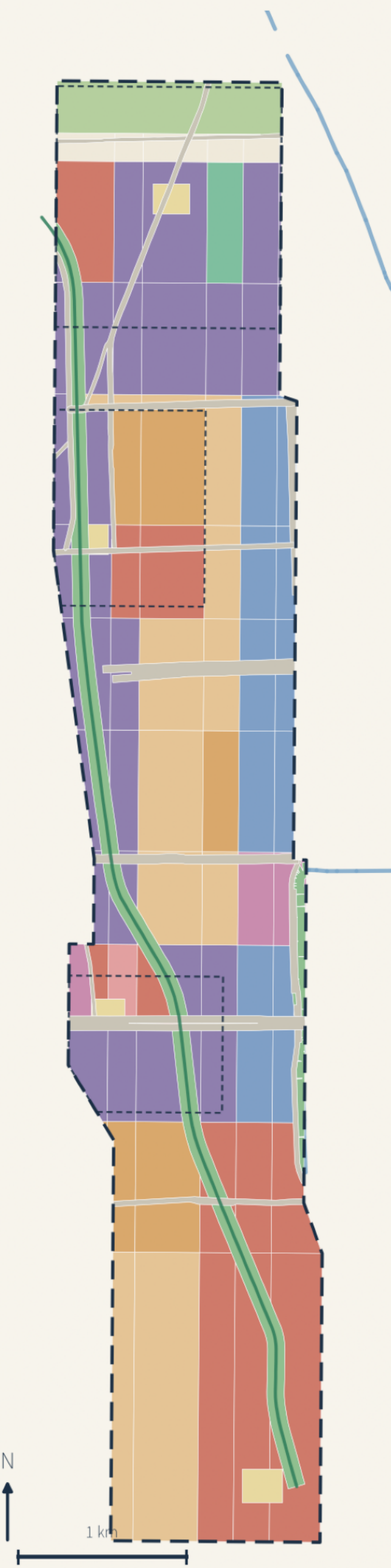
The Two-Way Corridor (concept)



- Surface Line • 7.8 km heritage slow spine
- Under Line • Qinghuayuan HSR tunnel (indicative)
- Design area (provisional)
- Key areas (provisional)
- Crossings (8 east-west stitches)
- Qinghe / Xiaoyue R.
- Metro Line 13

## Three-Level Scope & Land Use

43.6 km<sup>2</sup> research • 11.4 km<sup>2</sup> design • 368.4 ha key areas — 189-parcel full partition



|                      |                                              |
|----------------------|----------------------------------------------|
| Research             | 258.0 × 10 <sup>4</sup> m <sup>2</sup> 22.0% |
| Housing              | 209.3 × 10 <sup>4</sup> m <sup>2</sup> 18.4% |
| Commercial           | 203.5 × 10 <sup>4</sup> m <sup>2</sup> 17.9% |
| Education            | 111.5 × 10 <sup>4</sup> m <sup>2</sup> 9.8%  |
| Community svc.       | 89.9 × 10 <sup>4</sup> m <sup>2</sup> 7.9%   |
| Park green           | 80.3 × 10 <sup>4</sup> m <sup>2</sup> 7.0%   |
| Roads                | 74.1 × 10 <sup>4</sup> m <sup>2</sup> 6.5%   |
| Protective green     | 40.6 × 10 <sup>4</sup> m <sup>2</sup> 3.6%   |
| Culture              | 20.8 × 10 <sup>4</sup> m <sup>2</sup> 1.8%   |
| White land (reserve) | 18.6 × 10 <sup>4</sup> m <sup>2</sup> 1.6%   |
| Sports               | 15.4 × 10 <sup>4</sup> m <sup>2</sup> 1.4%   |
| Plaza                | 11.7 × 10 <sup>4</sup> m <sup>2</sup> 1.0%   |
| Health               | 6.3 × 10 <sup>4</sup> m <sup>2</sup> 0.6%    |

## Core thesis

Read 'Jing-Zhang' in full — the corridor has two ends. In 1909 the railway carried energy and engineering confidence into Beijing; in 2019 the HSR kept racing beneath the heritage park through Qinghuayuan Tunnel; in 2022 'Eastern Data, Western Computing' placed training compute on Zhangjiakou's green power. The belt is not a 9-km monologue but the southern terminal of a 200-km two-way relationship: power and compute up-line, models, talent, and scenarios down-line.

Structure • Surface+Under Lines / 3 Stations 1 Gate / 2 Platforms 8 Crossings / Cooling Loop

Surface Line = the 7.8-km heritage slow spine for people and culture; Under Line = the HSR in the tunnel for speed and reach. Three stations = Zhongzhi (training & standards), Origin (campus-city switch-back), Bell (content & data assets); one gate = the Two-Way Gate at Beijing North. West Platform = Zhongguancun services; East Platform = Xiaoyue scenarios; eight crossings stitch east and west. The Twin-River Cooling Loop joins the Qinghe and Xiaoyue with cabin waste-heat reuse.

Institutional core • the Two-Way Protocol

Every AI deployment is a paired run: an up-line run that collects data and consumes compute must be matched by a down-line run delivering services and public returns; who can stop it and how to exit is written on every card. Extraction-only deployments are never scheduled. Releases follow an annual public Timetable.

Three key areas • the density gradient from north to south

| Key area                     | Area     | Buildings | Concept GFA            | Concept gross PR | Leading land use | Character |
|------------------------------|----------|-----------|------------------------|------------------|------------------|-----------|
| Zhongzhi • Zhongzhiyuan      | 192.1 ha | 47        | 1.448 M m <sup>2</sup> | 0.75             | research 43.1%   | campus    |
| Origin • AI Origin Community | 104.3 ha | 49        | 1.005 M m <sup>2</sup> | 0.96             | community 35.7%  | district  |
| Bell • Dazhongsi             | 72.0 ha  | 38        | 1.208 M m <sup>2</sup> | 1.68             | research 59.3%   | urban     |

Concept gross plot ratio is concept floor area divided by key-area area — not a statutory plot ratio. Land use is recomputed parcel-by-parcel against each key area in EPSG:4548 and covers each area exactly.

Two scope facts • disclosed by us

(1) Of the eight crossings, Qinghe Bay sits about 85 m beyond the northern edge of the overall design area — its job is to carry the Surface Line past the boundary to the Qinghe waterfront, so crossing is functionally necessary and its land is excluded from in-area recomputation. (2) Qinghuayuan Station is about 389 m west of the area, inside the coordinated research scope, so run S8 is a deliberately cross-scope ceremonial venue. Both were found by our own recomputation and registered in assumptions.json.

11.40 km<sup>2</sup> design area (provisional, +0.0006%)

368.4 ha three key areas (±0.01%)

7.8 km through slow spine

8+3+1 12 scenario nodes

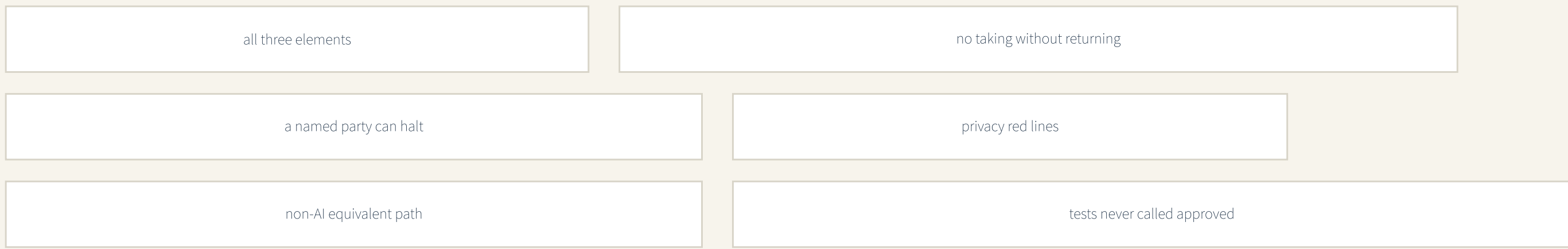
12/3 cards / industry tests

14 renewal projects, 3 phases



Two-Way Protocol • a recomputable admission contract

Six rules taken line by line from this proposal, encoded as a machine-readable contract and a tabletop exercise; reviewers can run it themselves rather than take the text on trust.



12 / 12 schedulable

Twelve service cards

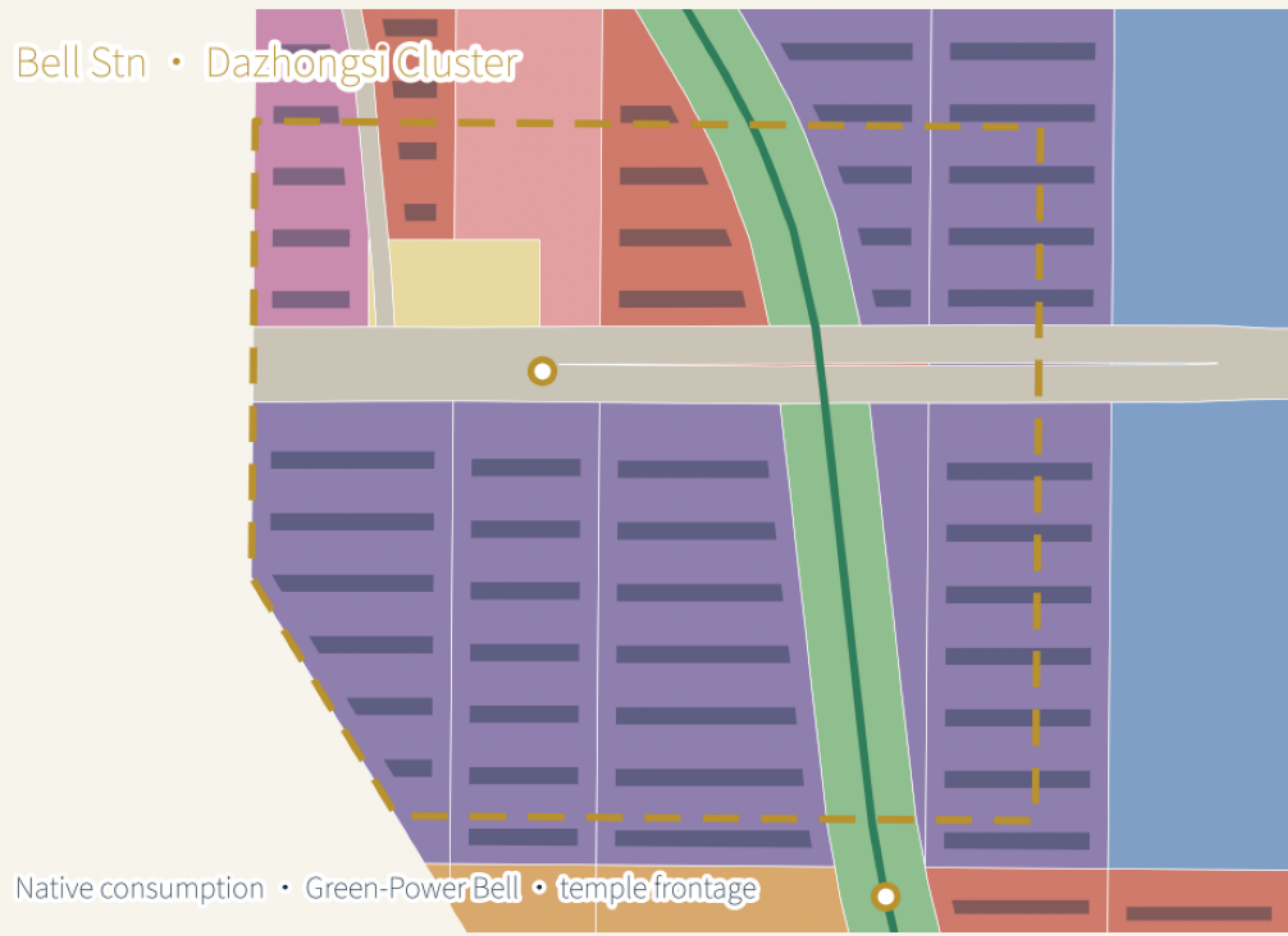
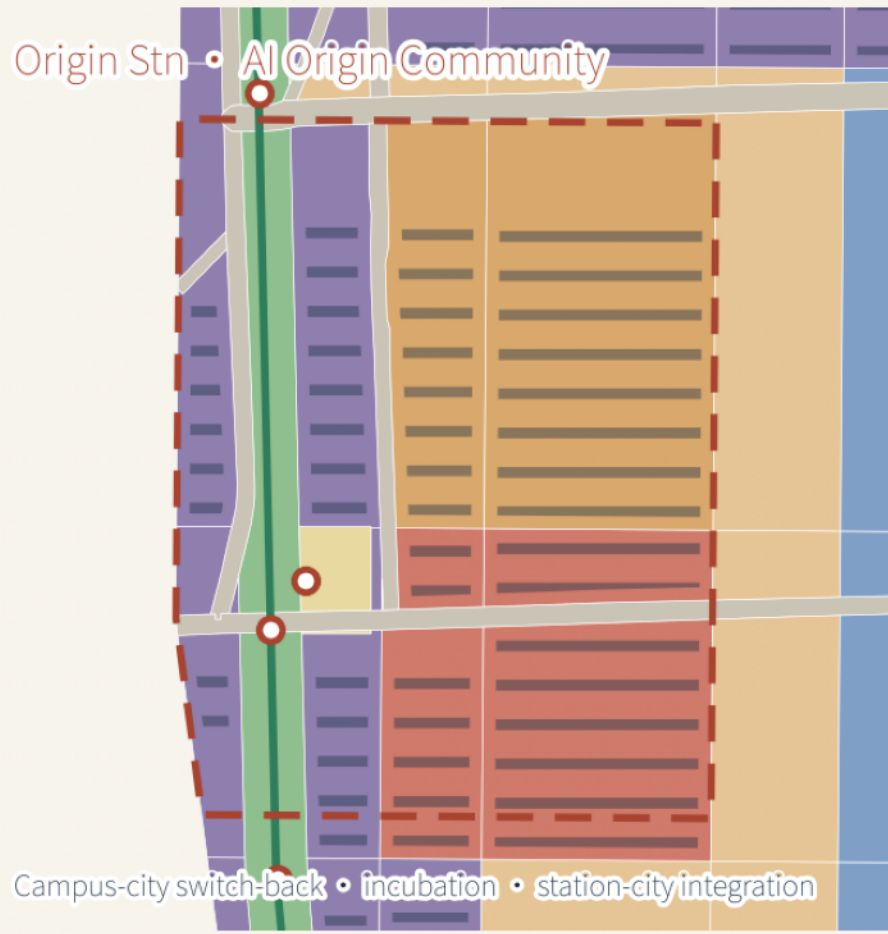
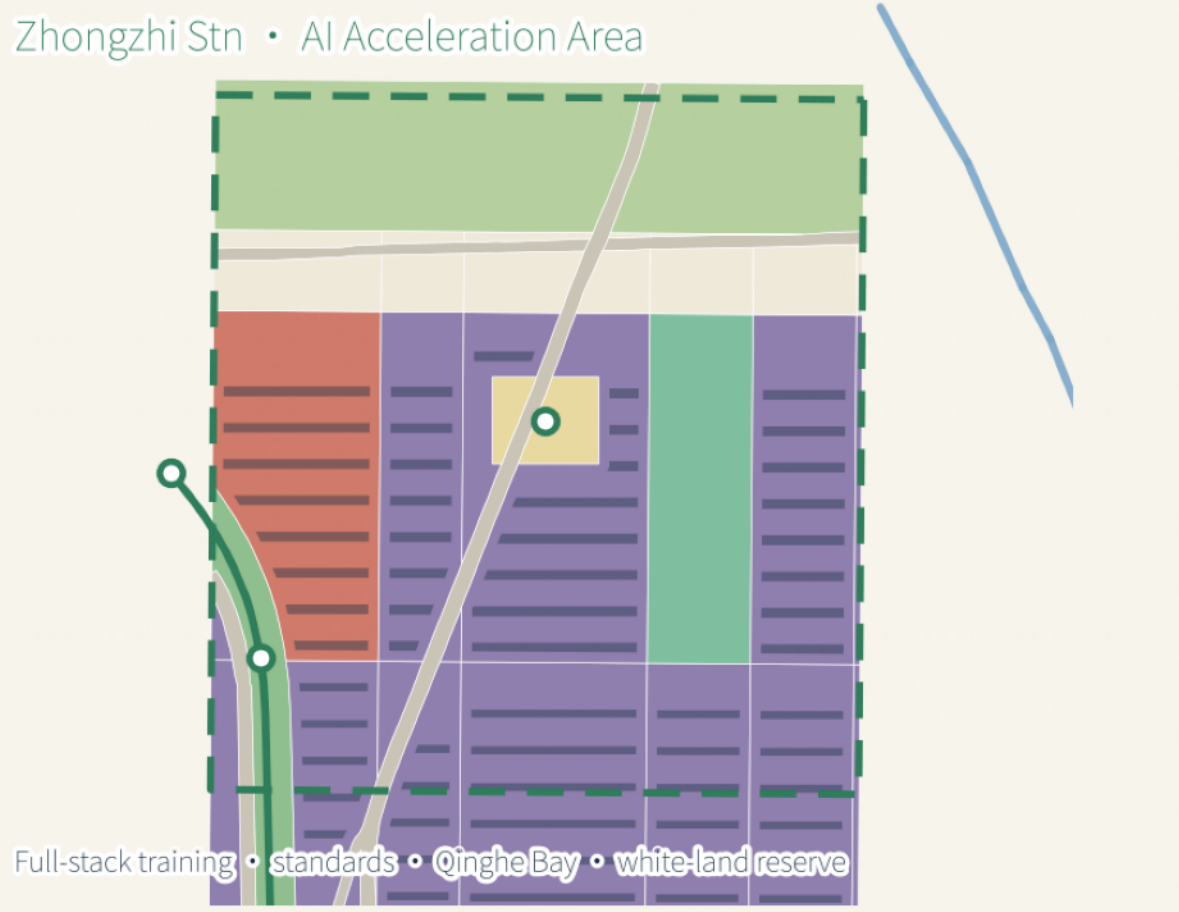
6 / 6 caught by their rule

Deliberately broken runs

node visual/assets/run\_twoway\_tabletop.js • Node 18+, zero dependencies

Detailed Design of Three Key Areas

Zhongzhi (192.1 ha) • Origin (104.3 ha) • Bell (72.0 ha) — implementation-scheme depth (concept)



Boundary status: provisional rough frame, self-derived from OSM centerlines of announced streets and area-calibrated; not an official redline; no approval/ownership precise area use; full recalculation when official polygons arrive.  
Base data © OpenStreetMap contributors (ODBL). Derived from packaged GeoJSON + metrics, concept proposal, w943 / jingzhang two-way line • 2020-08

Mobility & Blue-Green Systems

14 micro-circulation streets • through spine • 8 crossings • cooling loop • 4 station hubs

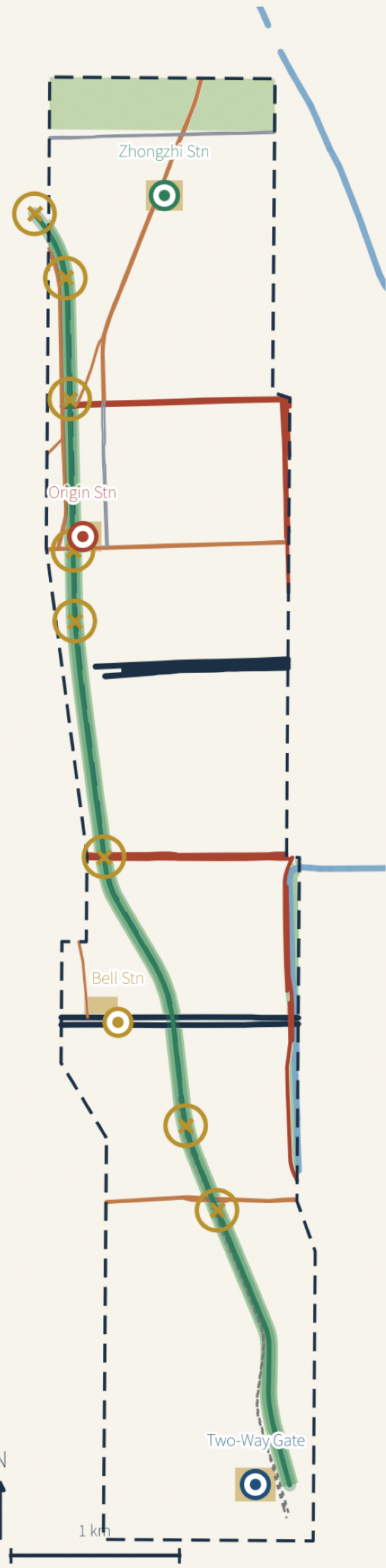
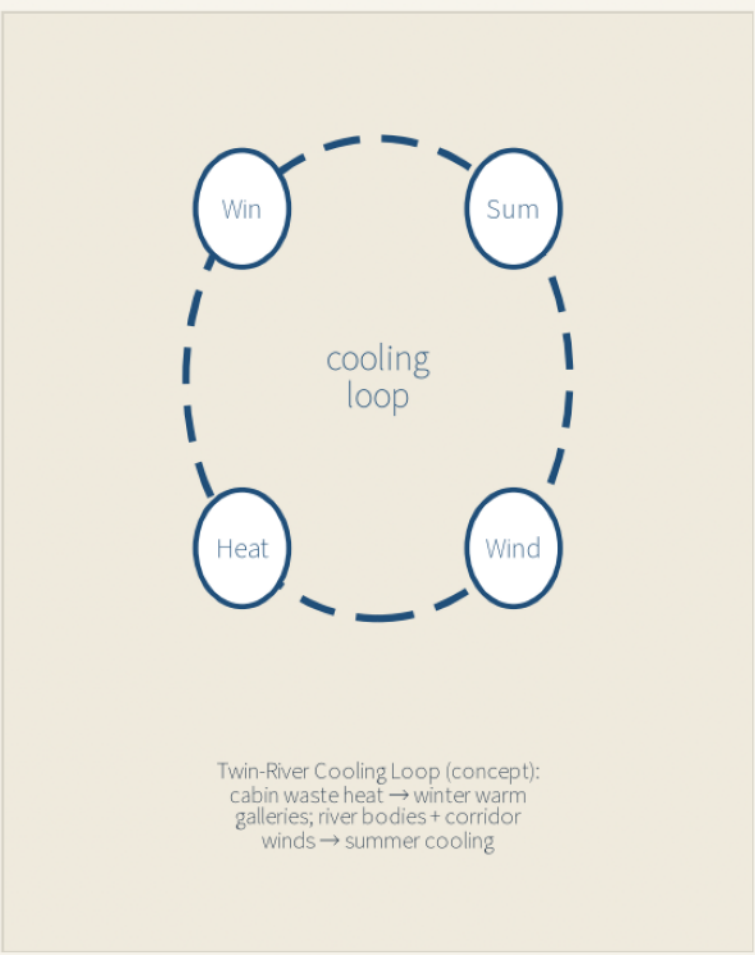
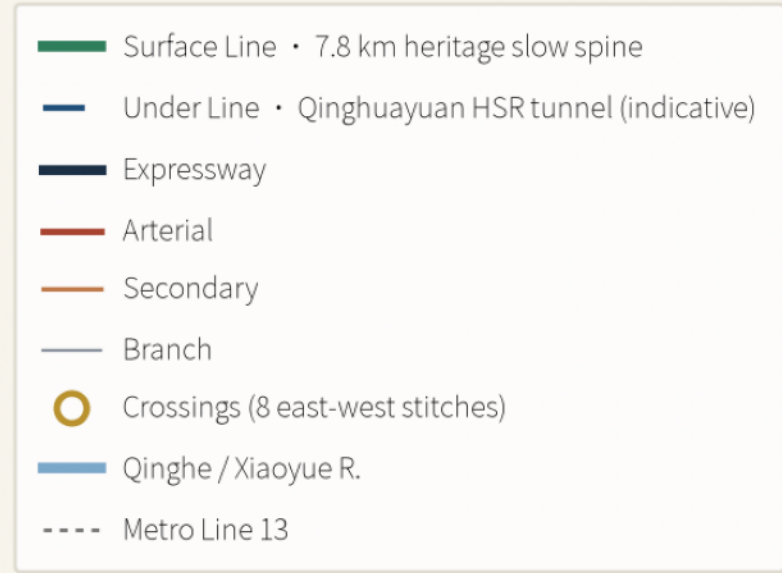


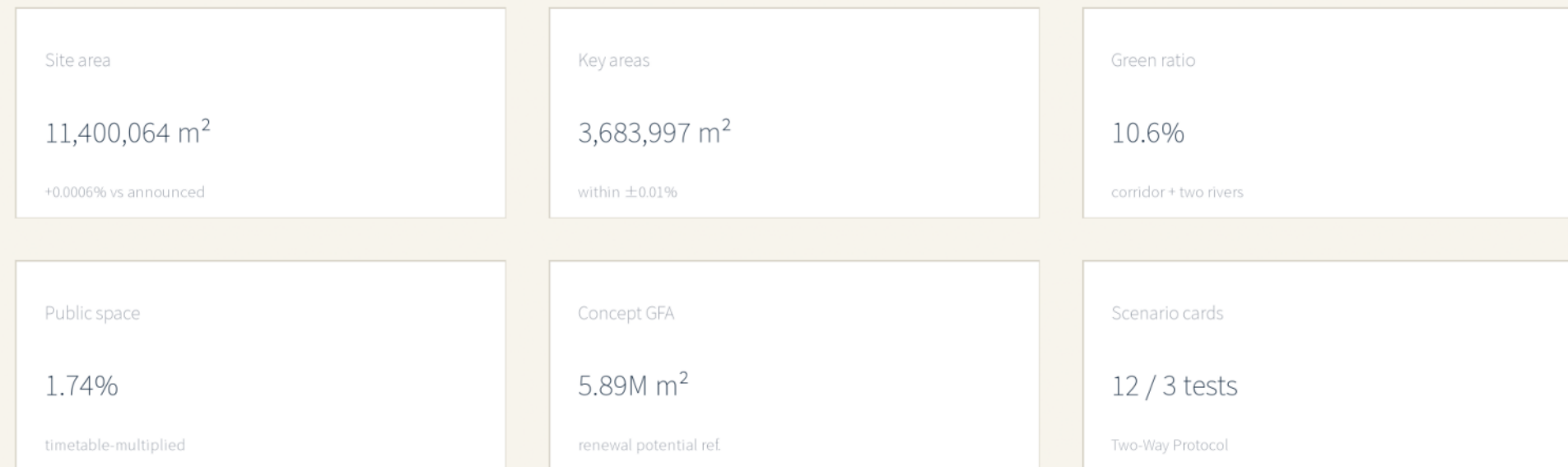
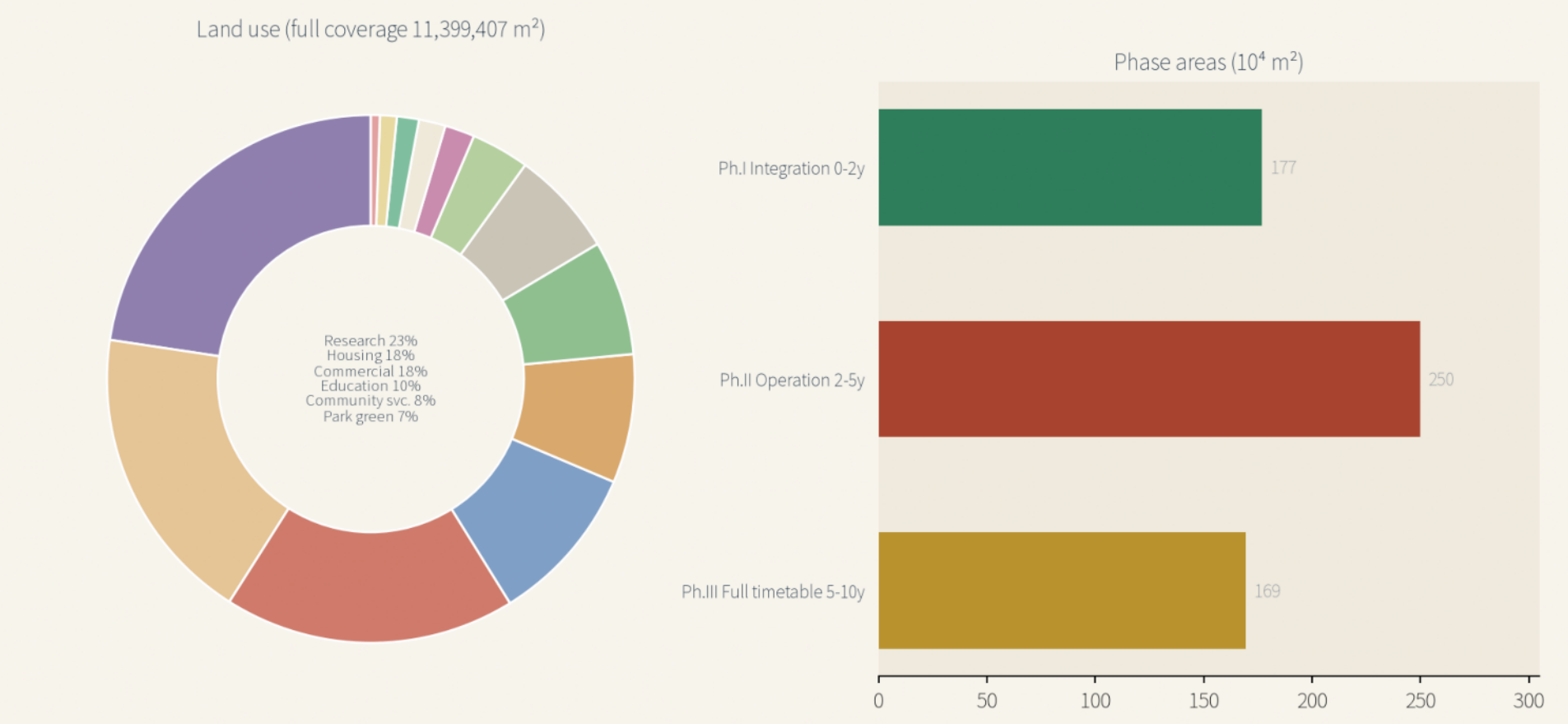
FIG-04



Boundary status: provisional rough frame, self-derived from OSM centerlines of announced streets and area-calibrated; not an official redline; no approval/ownership precise area use; full recalculation when official polygons arrive.  
Base data © OpenStreetMap contributors (ODBL). Derived from packaged GeoJSON + metrics, concept proposal, w943 / jingzhang two-way line • 2020-08

Metrics Recalculation & Evidence Chain

All spatial metrics recomputed from packaged GeoJSON in EPSG:4548 • spatial review PASS



Evidence chain: every claim traces to layers and metrics



Pending official data (registered, never faked): FAR • max height • AI index baseline • talent density baseline

Boundary status: provisional rough frame, self-derived from OSM centerlines of announced streets and area-calibrated; not an official redline; no approval/ownership precise area use; full recalculation when official polygons arrive.  
Base data © OpenStreetMap contributors (ODBL). Derived from packaged GeoJSON + metrics, concept proposal, w943 / jingzhang two-way line • 2020-08

Twelve run-cards (Two-Way Protocol: what goes up • what comes down • who can stop it)

| Run | Scenario                | Node         | Up-line            | Down-line           | Review & exit    |
|-----|-------------------------|--------------|--------------------|---------------------|------------------|
| T1  | Low-speed delivery test | Qinghe Bay   | de-identified ops  | free delivery hours | on-site stop     |
| T2  | Traffic dispatch test   | crossings    | anonymous flows    | published waits     | manual fallback  |
| T3  | Copilot sandbox         | Bell Stn     | authorized tickets | SME quota           | human switchable |
| S4  | Cultural guide          | mileposts    | anonymous heat     | 4-language guiding  | human review     |
| S5  | Health navigation       | Xueyuan Rd   | voluntary asks     | elder escort        | physician review |
| S6  | Safety ops review       | the Gate     | incident data      | published drills    | post-hoc only    |
| S7  | Waste-heat dispatch     | cooling loop | telemetry          | warm gallery        | manual bypass    |
| S8  | Model release day       | release hall | registration       | open weights        | community review |
| S9  | Accessible walking      | 8 crossings  | facility status    | blind guidance      | hotline backstop |
| S10 | Campus matching         | Origin hall  | licensed list      | shared returns      | manager review   |
| S11 | Micro-climate           | whole spine  | sensors            | heat warnings       | stoppage points  |
| S12 | Green power chime       | Bell plaza   | power share        | hourly chime        | auditable source |